

Haocheng He

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PROFESSIONAL SUMMARY

Results-driven engineering graduate with a strong foundation in operations research, data analytics, and simulation modelling. Experienced across aviation operations, supply-chain optimisation, and technology risk through four industry internships. Holds multiple invention patents and software copyrights. Seeking roles in Operations Analytics and Technology Risk.

EDUCATION

The Hong Kong Polytechnic University (PolyU) Hong Kong SAR
Master of Engineering in Aviation Engineering Sep 2024 – Present
CGPA: 3.95 / 4.30

Guangdong University of Technology (GDUT) Guangzhou, China
Bachelor of Engineering in Industrial Engineering Sep 2020 – Jun 2024
GPA: 3.67 / 4.0 (Top 20%)

Minor in Law Sep 2021 – Jun 2024
GPA: 3.61 / 4.0

INTERNSHIP EXPERIENCE

Operations Research Intern Jun 2023 – Sep 2023
Supply-Chain Analytics Co. Guangzhou, China

- Built an optimisation model using OR-Tools to redesign warehouse picking routes, reducing average travel distance by 18% across 3 facilities.
- Automated weekly KPI dashboards in Python (pandas + matplotlib), cutting manual reporting time from 4 hours to 20 minutes per cycle.
- Analysed 150k+ shipment records to identify bottlenecks, proposing layout changes that improved throughput by 12%.

Aviation Data Intern Jan 2023 – May 2023
Airport Authority Partner Project Hong Kong SAR

- Developed a discrete-event simulation in AnyLogic modelling passenger flows for a terminal redesign, testing 5 layout scenarios.
- Processed 2M+ flight-movement records with SQL to build a delay-prediction feature set (accuracy 82%).
- Collaborated with a 4-person team to deliver a feasibility report adopted by the project sponsor.

Manufacturing Simulation Intern Jul 2022 – Oct 2022
Automotive Parts Manufacturer Guangzhou, China

- Created a Plant Simulation model of a 12-station assembly line, identifying a bottleneck station responsible for 23% idle time.
- Proposed a revised shift schedule that increased utilisation from 71% to 84% without additional head-count.
- Documented simulation methodology in a 30-page technical report for the client's continuous-improvement team.

Business Intelligence Intern Jun 2021 – Sep 2021
Technology Consulting Firm Shenzhen, China

- Designed and populated a MySQL data warehouse consolidating 4 legacy systems, serving 15+ concurrent analyst queries.
- Built interactive Tableau dashboards tracking revenue, cost, and margin KPIs for C-suite stakeholders.
- Wrote Python ETL scripts that reduced data-pipeline runtime by 40% compared to the previous manual process.

SKILLS

Programming: Python, Java, SQL

Tools & Platforms: AnyLogic, Plant Simulation, OR-Tools, MySQL, Tableau, Git

Languages: English (IELTS 6.5), Mandarin (native), Cantonese (conversational)

PUBLICATIONS & INTELLECTUAL PROPERTY

- 3 Invention Patents (granted) – optimisation algorithms for logistics and scheduling.
- 1 Utility Model Patent – warehouse equipment ergonomic design.
- 2 Software Copyrights – simulation toolkits for manufacturing line analysis.
- Outstanding Undergraduate Thesis Award – “Simulation-Based Optimisation of Mixed-Model Assembly Lines.”